

Name: _____

Teacher: _____

SYDNEY TECHNICAL HIGH SCHOOL



Mathematics Extension 1

Preliminary Yearly Examination September 2009

Time allowed — 90 minutes

Instructions

- Reading time — 5 minutes
- Approved calculators may be used.
- All necessary working must be shown. Marks may not be awarded for careless or badly arranged work.
- Marks awarded are shown on each question.
- Total marks — 60
- Attempt all questions.
- Start each question on a new page.

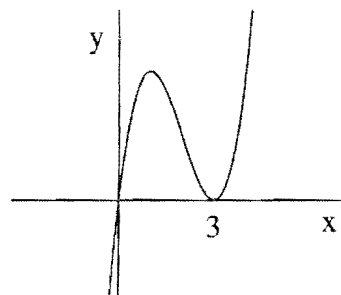
Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	Total

Question 1**Marks 10**

a) Use the Remainder Theorem to find the remainder when $P(x) = 2x^5 - 3x^2 + 7$ is divided by $(x - 3)$. (1)

b) Find the coordinates of the point which divides the interval $A(-2,1)$ to $B(3,2)$ in the ratio $4:3$. (2)

c) The sketch shows the cubic $y = P(x)$.



(i) What is the equation of $P(x)$ if it is monic? (2)

(ii) If it is not monic, find the equation of $P(x)$ if $P(2) = 4$? (2)

d) Find the equation of the normal to the curve $y = x^{\frac{3}{2}}$ at the point $(4, 8)$. (3)

Question 2**Marks 10**

a) Find the acute angle between the lines $2x + y - 3 = 0$ and $5x - 2y + 4 = 0$ to the nearest minute. (2)

b) The cubic $3x^3 - 8x^2 + 6x - 1$ has roots α , β and γ .

Evaluate: (i) $\alpha + \beta + \gamma$ (1)

(ii) $\alpha\beta\gamma$ (1)

(iii) $\alpha\beta + \alpha\gamma + \beta\gamma$ (1)

(iv) $\left(\alpha\beta + \frac{1}{\alpha\beta}\right) + \left(\alpha\gamma + \frac{1}{\alpha\gamma}\right) + \left(\beta\gamma + \frac{1}{\beta\gamma}\right)$ (3)

c) Describe, with the aid of diagrams and mathematical language, the meaning of:

(i) *continuity* at a point on a curve. (1)

(ii) *differentiability* at a point on a curve. (1)

Question 3**Marks 10**

- a) Find the derivative of $y = x^2\sqrt{1-x}$, expressing your answer as a single fraction. (3)
- b) Solve $\frac{x-2}{x+2} \geq 1$ (3)
- c) (i) Find the distance of $R(1,-2)$ from the line $x - 2y + 3 = 0$. (2)
- (ii) Show, without graphing, that $R(1,-2)$ and $S(-2,3)$ are on opposite sides of the line $x - 2y + 3 = 0$. (1)
- d) Find the Cartesian equation represented by $x = \frac{t}{2}$ and $y = 2t^2$. (1)

Question 4**Marks 10**

- a) $f(x) = \sqrt{x^2 - 9}$
- What is: (i) the domain? (2)
- (ii) the range? (1)
- b) (i) Show that $\frac{1 + \cos 2A}{\sin 2A} = \cot A$ (2)
- (ii) Hence find the exact value of $\cot 22^\circ 30'$. (1)
- c) $P(x,y)$ is constrained such that $\angle APB = 90^\circ$ where A is $(1,-1)$ and B is $(7,3)$.
- (i) Find an expression for the gradient of AP . (1)
- (ii) Show that the locus of $P(x,y)$ is a circle. (2)
- (iii) Find its centre and radius. (1)

Question 5**Marks 10**

- a) A parabola has the equation $6y = x^2 - 4x - 14$.
- (i) Express the equation of the parabola in the form $(x - h)^2 = 4a(y - k)$. (2)
- Find:
- (ii) the coordinates of the vertex. (1)
- (iii) the coordinates of the focus. (1)
- (iv) the equation of the directrix. (1)
- b) Solve $3\sin x - 2\cos x = 2$, for $0^\circ \leq x \leq 360^\circ$. (3)
- c) Show that $\sqrt{x} + \frac{5}{\sqrt{x}} = 3$ has no real solutions. (2)

Question 6**Marks 10**

- a) Solve $\cos^2 \theta - \sin 2\theta = 0$, for $0^\circ \leq \theta \leq 360^\circ$. (3)
- b) $P(8p, 4p^2)$ and $Q(8q, 4q^2)$ are variable points on the parabola $x^2 = 16y$.
The chord PQ produced passes through the fixed point $(4, 0)$.
The tangents at P and Q meet at R .
- (i) Find the equation of the chord PQ . (1)
- (ii) Show that $p + q = 2pq$. (2)
- (iii) Find the coordinates of R in terms of p and q . (2)
- (iv) Find the equation of the locus of R . (2)

2009 Preliminary Extension 1 Mathematics Yearly - Solutions

Q1 a) $P(x) = 2x^5 - 3x^2 + 7$
 $P(3) = 2 \times 3^5 - 3 \times 3^2 + 7$
 $= 486 - 27 + 7$
 $= 466$
 $\therefore P(x) = (x-3) \cdot Q(x) + 466$

b) $A(-2, 1) \quad B(3, 2) \quad 4:3 = m:n$
 $\therefore x = \frac{n x_1 + m x_2}{m+n} \quad y = \frac{n y_1 + m y_2}{m+n}$
 $= \frac{3 \times -2 + 4 \times 3}{7} \quad = \frac{3 \times 1 + 4 \times 2}{7}$
 $= \frac{6}{7} \quad = \frac{11}{7}$

c) (i) equation $\Rightarrow y = x(x-3)^2$
 $= x(x^2 - 6x + 9)$
 $= x^3 - 6x^2 + 9x$

(ii) $y = a(x^3 - 6x^2 + 9x)$
 $\therefore 4 = a(8 - 24 + 18)$
 $= 2a$
 $\therefore a = 2$
 $\therefore y = 2x^3 - 12x^2 + 18x$

d) $y = x^{\frac{3}{2}}$
 $\therefore y = \frac{3}{2} x^{\frac{1}{2}}$
 $= \frac{3}{2} \sqrt{x}$
 $= 3 \text{ when } x = 4 \text{ (positive root as } y=8)$

$\therefore y - 8 = -\frac{1}{3}(x - 4)$

$\therefore 3y - 24 = 4 - x$
 $\therefore x + 3y - 28 = 0$

Q2 a) $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$
 $= \left| \frac{-2 - \frac{5}{2}}{1 + -2 \cdot \frac{5}{2}} \right|$
 $= \left| \frac{-4\frac{1}{2}}{-4} \right|$
 $= \left| \frac{9}{8} \right|$

$\therefore \theta = \tan^{-1}\left(\frac{9}{8}\right)$
 $= 48^\circ 22'$

$2x + y - 3 = 0$
 $m_1 = -2$

$5x - 2y + 4 = 0$
 $m_2 = \frac{5}{2}$

b) $3x^3 - 8x^2 + 6x - 1$

(i) $\alpha + \beta + \gamma = -\frac{b}{a}$
 $= \frac{8}{3}$

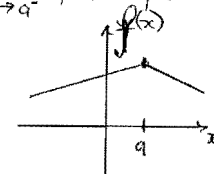
(ii) $\alpha\beta\gamma = -\frac{d}{a}$
 $= \frac{1}{3} \quad = \frac{c}{a}$
 $= 2$

(iv) $\left(\alpha\beta + \frac{1}{\alpha\beta}\right) + \left(\alpha\gamma + \frac{1}{\alpha\gamma}\right) + \left(\beta\gamma + \frac{1}{\beta\gamma}\right)$
 $= \alpha\beta + \alpha\gamma + \beta\gamma + \left(\frac{1}{\alpha\beta} + \frac{1}{\alpha\gamma} + \frac{1}{\beta\gamma}\right)$
 $= 2 + \frac{\alpha + \beta + \gamma}{\alpha\beta\gamma}$
 $= 2 + \frac{8}{3} \times 3$
 $= 10$

c) (i) continuity at $x=a$...

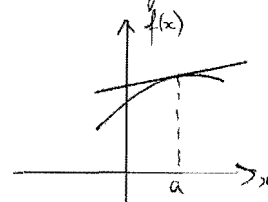
the limit from the left = value of fn
 $=$ limit from the right.

or $\lim_{x \rightarrow a^-} f(x) = f(a) = \lim_{x \rightarrow a^+} f(x)$



(ii) differentiability at $x=a$...

It is possible to find a derivative at $x=a$
or it is possible to draw a tangent to the curve at $x=a$.



example above is continuous but NOT differentiable at $x=a$.

Q3 a) $y = x^2 \sqrt{1-x}$
 $= x^2 (1-x)^{\frac{1}{2}}$

$\therefore y' = vu' + uv'$
 $= \sqrt{1-x} \cdot 2x + x^2 \cdot \frac{-1}{2\sqrt{1-x}}$
 $= 2x\sqrt{1-x} - \frac{x^2}{2\sqrt{1-x}}$

$= \frac{4x(1-x) - x^2}{2\sqrt{1-x}}$

$= \frac{4x - 4x^2 - x^2}{2\sqrt{1-x}}$

$= \frac{4x - 5x^2}{2\sqrt{1-x}}$

$= \frac{x(4-5x)}{2\sqrt{1-x}}$

Let $u = x^2$

$\therefore u' = 2x$

$v = (1-x)^{\frac{1}{2}}$

$\therefore v' = -\frac{1}{2}(1-x)^{-\frac{1}{2}}$

$= \frac{-1}{2\sqrt{1-x}}$

b) $\frac{x-2}{x+2} \geq 1$

NB $x \neq -2$

$\therefore (x-2)(x+2) \geq (x+2)^2$

$\therefore x^2 - 4 \geq x^2 + 4x + 4$

$\therefore 4x + 8 \leq 0$

$\therefore x + 2 \leq 0$

$\therefore x \leq -2$

but $x \neq -2$

$\therefore x < -2$

c) (i) $R(1, -2) \quad x - 2y + 3 = 0$

$d_1 = \left| \frac{Ax_1 + By_1 + C}{\sqrt{A^2 + B^2}} \right|$

$= \left| \frac{1 \times 1 - 2 \times -2 + 3}{\sqrt{1 + 4}} \right|$

$= \left| \frac{8}{\sqrt{5}} \right|$

$= \frac{8}{\sqrt{5}} \text{ units}$

(ii) $S(-2, 3)$

$d_2 = \left| \frac{1 \times -2 - 2 \times 3 + 3}{\sqrt{5}} \right|$

$= \left| \frac{-5}{\sqrt{5}} \right|$

$\therefore S$ is on opposite side of line
 $\therefore$ values inside absolute value
 signs are opposite in sign.

d) $x = \frac{t}{2} \quad y = 2t^2$

$\therefore t = 2x$

$\therefore y = 2(2x)^2$
 $= 8x^2$

or $x^2 = \frac{y}{8}$

Q4 a) $f(x) = \sqrt{x^2 - 9}$

(i) $f(x)$ is undefined when $x^2 - 9 < 0$
 $x^2 < 9$

$-3 < x < 3$

$\therefore \text{domain} \Rightarrow \{x : x \leq -3, x \geq 3\}$

(ii) $\sqrt{x^2 - 9} \geq 0$

$\therefore \text{range} \Rightarrow \{y : y \geq 0\}$

b) (i) RTS $\frac{1 + \cos 2A}{\sin 2A} = \cot A$

LHS = $\frac{1 + \cos 2A}{\sin 2A}$

$= \frac{1 + \cos^2 A - \sin^2 A}{2 \sin A \cos A}$

$= \frac{\cos^2 A + \cos^2 A}{2 \sin A \cos A}$

$= \frac{2 \cos^2 A}{2 \sin A \cos A}$

$= \frac{\cos A}{\sin A}$

$= \cot A$

$= \text{RHS} \quad \text{QED}$

NB Use $\Rightarrow$
 $\cos 2A = 2\cos^2 A - 1$
 instead. It is
 better.

(ii) $\cot 22.5^\circ = \frac{1 + \cos 45^\circ}{\sin 45^\circ}$

$= \frac{1 + \frac{1}{\sqrt{2}}}{\frac{1}{\sqrt{2}}}$

$= \frac{\sqrt{2} + 1}{1} \times \sqrt{2}$

$= \sqrt{2} + 1$

c) $A(1, -1) \quad B(7, 3) \quad \angle APB = 90^\circ$

(i) grad AP = $\frac{y_2 - y_1}{x_2 - x_1}$
 $= \frac{y - (-1)}{x - 1}$
 $= \frac{y + 1}{x - 1}$

(ii) grad BP = $\frac{y - 3}{x - 7}$

Now $\frac{y + 1}{x - 1} \times \frac{y - 3}{x - 7} = -1$

$\therefore (y + 1)(y - 3) = -(x - 1)(x - 7)$

$\therefore y^2 - 2y - 3 = -x^2 + 8x - 7$

$\therefore (x - 4)^2 + (y - 1)^2 = 13$ which is a circle.

(iii) centre $\Rightarrow (4, 1)$
 radius $\Rightarrow \sqrt{13}$ units

Q5

$$a) 6y = x^2 - 4x - 14$$

$$(i) x^2 - 4x - 14 = 6y$$

$$\therefore (x-2)^2 - 18 = 6y$$

$$\therefore (x-2)^2 = 6y + 18$$

$$= 6(y+3)$$

$$\therefore (x-2)^2 = 4 \times \frac{3}{2} (y+3)$$

$$(ii) \text{ vertex } \Rightarrow (2, -3)$$

$$(iii) \text{ focus } \Rightarrow (2, -\frac{5}{2})$$

$$(iv) \text{ directrix } \Rightarrow y = -4\frac{1}{2}$$

$$b) 3\sin x - 2\cos x = 2$$

$$\therefore 3 \times \frac{2t}{1+t^2} - 2 \times \frac{(1-t^2)}{1+t^2} = 2$$

$$\therefore \frac{6t}{1+t^2} - \frac{2(1-t^2)}{1+t^2} = 2$$

$$\therefore 6t = 4$$

$$\therefore t = \frac{2}{3}$$

$$\therefore \tan \frac{\theta}{2} = \frac{2}{3}$$

$$\therefore \frac{\theta}{2} = 33^\circ 41' 24''$$

$$\therefore \theta = 67^\circ 23'$$

$$c) \sqrt{x} + \frac{5}{\sqrt{x}} = 3$$

$$\therefore x + 5 = 3\sqrt{x}$$

$$\therefore x^2 + 10x + 25 = 9x$$

$$\therefore x^2 + x + 25 = 0$$

$$\therefore x = \frac{-1 \pm \sqrt{1 - 4 \times 1 \times 25}}{2}$$

$$= \frac{-1 \pm \sqrt{-99}}{2}$$

$$\therefore \Delta = -99$$

$$< 0$$

$\therefore$ $\nexists$ real solutions.

Q6

$$a) \cos^2 \theta - \sin 2\theta = 0$$

$$\therefore \cos^2 \theta - 2\sin \theta \cos \theta = 0$$

$$\therefore \cos \theta (\cos \theta - 2\sin \theta) = 0$$

$$\therefore \cos \theta = 0 \text{ or } \cos \theta = 2\sin \theta$$

$$\therefore \theta = 90^\circ, 270^\circ \quad \therefore \frac{1}{2} = \frac{\sin \theta}{\cos \theta}$$

$$= \tan \theta$$

$$\therefore \theta = 26^\circ 34', 90^\circ, 206^\circ 34', 270^\circ \quad \therefore \theta = 26^\circ 34', 206^\circ 34'$$

$$b) P(8p, 4p^2) \quad Q(8q, 4q^2) \text{ on } x^2 = 16y$$

$$\therefore a = 4$$

$$a) \text{ chord } PQ \Rightarrow y = \left(\frac{p+q}{2}\right)x - apq$$

$$\therefore y = \left(\frac{p+q}{2}\right)x - 4pq$$

$$b) (4,0) \text{ satisfies chord}$$

$$\therefore 0 = 4\left(\frac{p+q}{2}\right) - 4pq$$

$$\therefore 4pq = 4\left(\frac{p+q}{2}\right)$$

$$\therefore pq = \frac{p+q}{2}$$

$$\therefore 2pq = p+q$$

$$c) \text{ tangents meet at } (a(p+q), apq) = (a \cdot 2pq, apq)$$

$$= (2apq, apq)$$

$$d) x = 2apq$$

$$y = apq$$

$$\therefore pq = \frac{y}{a}$$

$$\therefore x = 2ay$$

$$\therefore x = 2y \Rightarrow \text{ locus of } R \text{ (with } x \leq 0)$$